

## 1 Introduction

In this practice, we will work on fitting different volatility models (ARCH and GARCH). We will also focus on discussing the validation and selection of the model. As done in previous practices, each student group must follow the practice script below, performing the necessary calculations and graphs in the R environment. The obtained results and answers to the questions should be compiled into a document containing your solution, created, for example, in R Markdown or Microsoft Word. Please make sure to include your full names and group number at the beginning of the document. Submit the questionnaire solution in PDF format by uploading it to the platform under the corresponding task section before the deadline.

## 2 Questionnaire

In this questionnaire, we will work with the following time series:

- KO: The Coca-Cola Company (KO) stock (NYSE)

You can download the data using this code:

```
library(quantmod)
getSymbols("KO")
```

You must:

1. Download the time series. You may work with the complete dataset (from 2007 to the most recent trading session).
2. Create a graphical representation of the time series using the Adjusted Closing Price.
3. Compute the log-returns of the series. It is not necessary to model the mean (e.g., using an ARIMA model).
4. Using the `fGarch` package, fit the following models to the log-returns: an ARCH(1) model, an ARCH(3) model, and a GARCH(1,1) model.
5. Write the expression of each fitted model.
6. Analyze the diagnostic results and determine which of the three models provides the best fit.